

Q1.

$I \Rightarrow iESeS$

$\Rightarrow i = ESeS$

$E \Rightarrow EoG$

$E \Rightarrow EoGoG$

$E \Rightarrow GoGoG$

$G \Rightarrow x$

$\Rightarrow ixo = GoG$

$G \Rightarrow y$

$\Rightarrow ixoyoG$

$G \Rightarrow w$

$\Rightarrow ixoyow = SeS$

$S \Rightarrow dLb$

$L \Rightarrow sL$

$L \Rightarrow s$

$\Rightarrow ixoyowdssbe = S$

$S \Rightarrow s$

$\Rightarrow ixoyowdssbes$

Q2. The grammar rules that have Left Recursion are:

$E \Rightarrow EoG \mid G$, left recursion is present in $E \Rightarrow EoG$ where E can recurse into E

$L \Rightarrow s \mid sL$, left recursion is present in $L \Rightarrow sL$ where L can recurse into L

We can rewrite these grammar rules using a new non-terminal such as E' and L'

$E \Rightarrow GE'$

$E' \Rightarrow oGE' \mid \epsilon$

$L \Rightarrow sL'$

$L' \Rightarrow sL' \mid \epsilon$

Grammar without left recursion:

$I \Rightarrow iES \mid iESeS$

$E \Rightarrow GE'$

$E' \Rightarrow oGE' \mid \epsilon$

$G \Rightarrow x \mid y \mid z \mid w$

$L \Rightarrow sL'$

$L' \Rightarrow sL' \mid \epsilon$

Q3. For a top-down parser, we will create a function for each non-terminal in the grammar and each function will consume the input based on the first symbol it encounters then apply the correct production rule. We will analyze how each function should behave based on the first symbol in the input string. Each function will check the first symbol in the input and determine what comes next. The function will consume the correct portion of the input and call other functions recursively where it is necessary. We find symbols for which there is more than one possible action for 's', since $S \Rightarrow s$ OR $L \Rightarrow sL'$.

Function I

First symbol : i

If followed by ESeS, apply $I \Rightarrow iES$

If followed by ESeS, apply $I \Rightarrow iESeS$

Function E

First symbols: x, y, z, w (G)

Apply $E \Rightarrow GE'$

Function E'

First symbol: o

Apply $E' \Rightarrow oGE'$

First symbol not o:

Apply $E' \Rightarrow \epsilon$ (empty production)

Function G

First symbols: x, y, z, w

Choose the production:

$G \Rightarrow x | y | z | w$

Function S

First symbol: s

Apply $S \Rightarrow s$

First symbol: d

Apply $S \Rightarrow dLb$

Function L

First symbol: s

Apply $L \Rightarrow sL'$

Function for L'

First symbol: s

Apply $L' \Rightarrow sL'$

Any other symbol (or end of string):

Apply $L' \Rightarrow \epsilon$

Q4.

Left factoring for S:

$S \Rightarrow sS' \mid dLb$

$S' \Rightarrow \epsilon$

Left factoring for L:

$L \Rightarrow sL'$

$L' \Rightarrow L' \mid \epsilon$

Grammar after Left-Factored:

$I \Rightarrow iES \mid iESeS$

$E \Rightarrow GE'$

$E' \Rightarrow oGE' \mid \epsilon$

$G \Rightarrow x \mid y \mid z \mid w$

$S \Rightarrow sS' \mid dLb$

$S' \Rightarrow \epsilon$

$L \Rightarrow sL'$

$L' \Rightarrow L' \mid \epsilon$

Q5.

Function S

If first symbol s:

Apply $S \Rightarrow sS'$

If first symbol is d:

Apply $S \Rightarrow dLb$

Function S'

Apply $S' \Rightarrow \epsilon$

Function L

If first symbol is s:

Apply $L \Rightarrow sL'$

Function L'

If first symbol is s:

Apply $L' \Rightarrow sL'$

Otherwise:

Apply $L' \Rightarrow \epsilon$